

DSA0502 QUERY PROCESSING FOR DATASCIENCE

CO1-AT1-DATA INTERPRETATION

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61. Customer Data Cleaning

Aim

To clean and merge customer data using Python and SQLite.

Procedure

1. Connect to the SQLite database.
2. Create Customer and Region tables.
3. Insert and read the data.
4. Remove duplicates, handle missing values, and merge the tables.
5. Display the summary and save the cleaned data to a CSV file.

Program

```
import sqlite3

import pandas as pd

conn = sqlite3.connect("customer.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Customers(

CustomerID INTEGER PRIMARY KEY,

CustomerName TEXT,

RegionID INTEGER,

Age INTEGER)

""")

cursor.execute("""

CREATE TABLE IF NOT EXISTS Regions(

RegionID INTEGER PRIMARY KEY,

RegionName TEXT)
```

```

""")
cursor.execute("INSERT INTO Customers VALUES(1,'Sindhu',101,20)")
cursor.execute("INSERT INTO Customers VALUES(2,'Rahul',102,21)")
cursor.execute("INSERT INTO Customers VALUES(3,'Priya',101,22)")
cursor.execute("INSERT INTO Regions VALUES(101,'North')")
cursor.execute("INSERT INTO Regions VALUES(102,'South')")
conn.commit()

customers=pd.read_sql_query("SELECT * FROM Customers",conn)
regions=pd.read_sql_query("SELECT * FROM Regions",conn)
customers=customers.fillna("Unknown")
customers=customers.drop_duplicates()
merged=pd.merge(customers,regions,on="RegionID")
print(merged)
summary=merged.groupby("RegionName")["CustomerID"].count()
print(summary)
print("Highest Region:",summary.idxmax())
merged.to_csv("Cleaned_Customers.csv",index=False)
conn.close()

```

Output

```

)
   CustomerID  CustomerName  RegionID  Age  RegionName
0           1         Sindhu        101   20         North
1           2          Rahul        102   21         South
2           3          Priya        101   22         North
RegionName
North      2
South      1
Name: CustomerID, dtype: int64
Highest Region: North

```

Result

Customer data was cleaned, merged, and exported successfully.

62. Product Inventory Integration

Aim

To integrate and clean product inventory data using Python and SQLite.

Procedure

1. Connect to the SQLite database.
2. Create Product and Category tables.
3. Insert and read the data.
4. Remove duplicates, handle missing values, and merge the tables.
5. Display the summary and save the cleaned data to a CSV file.

Program

```
import sqlite3

import pandas as pd

conn = sqlite3.connect("product.db")

cursor = conn.cursor()

cursor.execute("""

CREATE TABLE IF NOT EXISTS Products(

ProductID INTEGER PRIMARY KEY,

ProductName TEXT,

CategoryID INTEGER,

Price REAL)

""")

cursor.execute("""

CREATE TABLE IF NOT EXISTS Categories(

CategoryID INTEGER PRIMARY KEY,

CategoryName TEXT)

""")

cursor.execute("INSERT INTO Products VALUES(1,'Laptop',201,50000)")

cursor.execute("INSERT INTO Products VALUES(2,'Mouse',202,500)")

cursor.execute("INSERT INTO Products VALUES(3,'Keyboard',202,1000)")
```

```

cursor.execute("INSERT INTO Categories VALUES(201,'Electronics')")
cursor.execute("INSERT INTO Categories VALUES(202,'Accessories')")
conn.commit()

products=pd.read_sql_query("SELECT * FROM Products",conn)
categories=pd.read_sql_query("SELECT * FROM Categories",conn)
products=products.fillna("Unknown")
products=products.drop_duplicates()
merged=pd.merge(products,categories,on="CategoryID")
print(merged)
summary=merged.groupby("CategoryName")["Price"].sum()
print(summary)
print("Highest Category:",summary.idxmax())
merged.to_csv("Cleaned_Products.csv",index=False)
conn.close()

```

Output

	ProductID	ProductName	CategoryID	Price	CategoryName
0	1	Laptop	201	50000.0	Electronics
1	2	Mouse	202	500.0	Accessories
2	3	Keyboard	202	1000.0	Accessories

CategoryName

Accessories	1500.0
Electronics	50000.0

Name: Price, dtype: float64
Highest Category: Electronics

Result

Product inventory data was integrated, cleaned, and exported successfully.